

L2.8 Antiderivatives

§3.9 — every derivative rule, read from right to left

THE ANTIDERIVATIVE TABLE

Given f' , an **antiderivative** is a function f whose derivative is f' .

given f'	an antiderivative	given f'	an antiderivative
$c g'(x)$	$c g(x)$	$\cos x$	$\sin x$
$g'(x) + h'(x)$	$g(x) + h(x)$	$\sin x$	$-\cos x$
$x^n, n \neq -1$	$\frac{x^{n+1}}{n+1}$	$\sec^2 x$	$\tan x$
		$\sec x \tan x$	$\sec x$

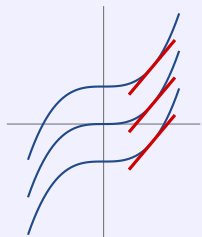
Rewrite first. Radicals, quotients and squares all become *powers* before any row applies.

$$\sqrt[3]{x} = x^{1/3}, \quad \frac{10}{x^9} = 10x^{-9}, \quad (x-5)^2 = x^2 - 10x + 25.$$

No reverse product rule, no reverse quotient rule. For anything the table does not reach: **guess, then differentiate to check.**

“ $n = -1$ is out because it divides by zero — no power of x has derivative $\frac{1}{x}$, though $\ln x$ does (Unit 5)”

The $+C$ family



all have same slope!

If f is an antiderivative of f' , so is $f + C$ — a constant differentiates to zero.

And there are **no others**. You proved that when you studied the **Mean Value Theorem**:

$$\boxed{f'(x) = g'(x) \text{ on an interval} \implies f(x) = g(x) + C}$$

So $+C$ is not a habit — it is the statement that the list is **complete**. Without that theorem there could be antiderivatives nobody has written down.

INITIAL VALUE PROBLEMS

An equation involving a derivative is a **differential equation**. The $+C$ answer is its *general* solution; a condition picks the *particular* one. Finding that C is an **initial value problem**.

1. Antidifferentiate, keeping the $+C$. **2.** Substitute the condition. **3.** Solve for the constant.

Repeat all three, in order, once per derivative. Given f'' you do it twice; given f''' , three times.

“one constant per antidifferentiation, and each one is pinned down before the next appears”

The order is the whole difficulty. Carry an unresolved C into the second antidifferentiation and it becomes $Cx + D$ — one equation, two unknowns.

POSITION FROM ACCELERATION

$a = v'$ and $v = x'$, so **position is the antiderivative of the antiderivative of acceleration**:
 $a(t) \longrightarrow v(t) \longrightarrow x(t)$.

It is the two-constant problem above in different letters — and one step is easier. Substituting $t = 0$ kills every term carrying a t , so

“ $v(0)$ is the constant — you read it off, you do not solve for it”

From a graph: where $v > 0$, x rises; where $v < 0$, x falls; where v crosses zero, x has a horizontal tangent and turns around. Near the ground, $a = -32 \text{ ft/s}^2$.

WATCH OUT

~~$\times$ $+C$ bookkeeping~~ $\rightarrow$ **it is the claim that the family is complete, and that claim is a theorem**

~~$\times$ the power rule backwards works for every $n \rightarrow n \neq -1$~~ , where the rule genuinely breaks.
No power of x has derivative $\frac{1}{x}$ — but a function does: it is $\ln x$, in Unit 5

~~$\times$ resolve both constants at the end~~ $\rightarrow$ **resolve the first before antidifferentiating again, or the second step carries $Cx + D$**

~~$\times$ antidifferentiate a product factor by factor~~ $\rightarrow$ **there is no reverse product rule and no reverse quotient rule. Guess, then differentiate to check**

~~$\times$ an antiderivative of $\sin x$ is $\cos x \rightarrow -\cos x$~~ . **The sign is the whole reason that row is in the table**

~~$\times$ $v(0)$ is something you solve for~~ $\rightarrow$ **$v(0)$ is the constant. Substituting $t = 0$ leaves C standing alone**

~~$\times$ divide by x , then antidifferentiate~~ $\rightarrow$ **rewrite as powers first. Division is not a rule that runs backwards; powers are**